Industrial tools

# Charts

Copy MCP Server

Analyze time series data, create calculations, set thresholds, monitor alerts, assess data quality, and visualize activities.

## Create a chart

1. **Navigate to Charts**

   Navigate to **Industrial tools** > **Charts**.

2. **Create a new chart**

   Select **+ New chart** to start a new chart. To build on an existing chart, select ⋮ to the right of the chart > **Duplicate**.

① **Toolbar** — hover over each tool to see a description. Use the **Slider** to view or compare time series values.

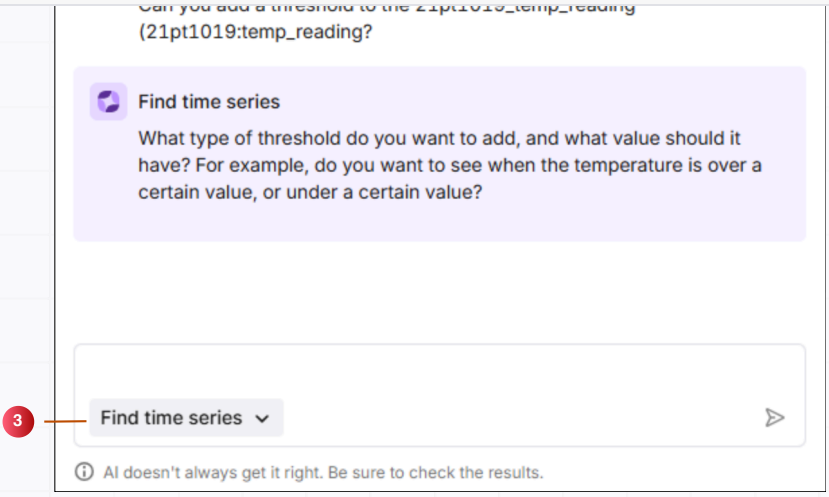
② **Open in...** — download, duplicate, delete, or **open** the chart in **Canvas**.

③ **Add data** — add a **time series**, or import or create a **calculation**.

④ **Agents** — open an **AI agent** to work with the chart.

⑤ **Sidebar** — set up **alerting**, **monitoring**, **thresholds**, or data **profiling**. Add **events** and **activities**.

## Use an agent to find and analyze data




① **New session** — start a new conversation with an agent.

② **Select locations** — see your current location. Select locations to choose which data the agent can access.

③ **Select agent** — select the agent you want to start a conversation with.

> To interact with an agent, you need these **capabilities**.
>
> Learn more about **Atlas AI**.

## Add a time series

To add a time series to your chart:

1. **Open Add time series**

   Select **Add data** > **Add time series**.

2. **Select and add time series**

line. Click ⓘ for more details.

## Add a calculation

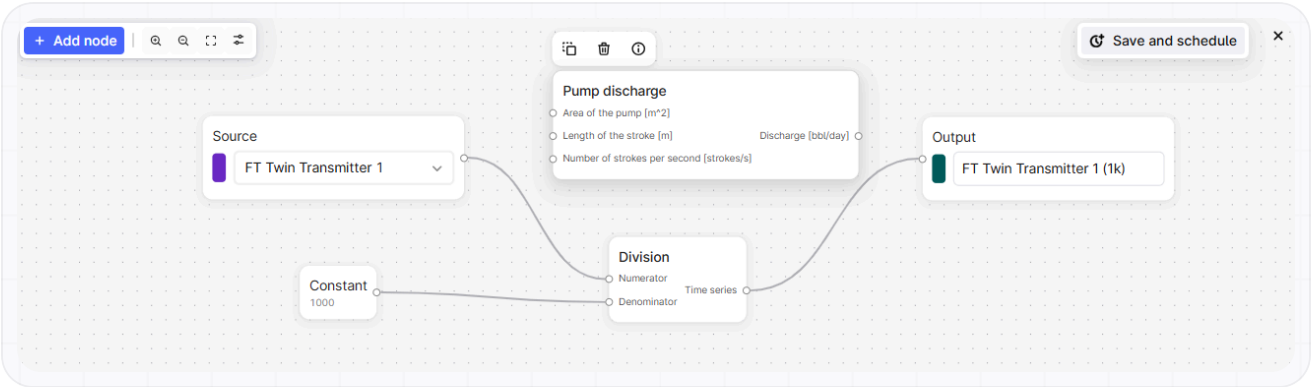
To add a calculation:

1. **Create a calculation**

   Select **Add data** > **Create calculation**.

2. **Add input sources**

   Select **Add node** or right-click in the panel below the chart to add input sources.




3. **Add constants or functions**

   Right-click the panel to add **constants** or **functions** to the calculation. Select ⓘ to view the function details or hover over the parameters for more information.

4. **Configure output**

   Pipe the calculations in order and include the final **output** node to display the result on the chart plot.

## Schedule a calculation

Schedule a calculation and save the result as a time series. You can also **monitor** the calculation.

1. **Save and schedule**

   **Add a calculation** and select **Save and Schedule**.

2. **Specify credentials**

   Specify the sign-in credentials for the schedule, and click **Next**. If you don't know the credentials, contact your CDF admin.

3. **Configure schedule settings**

   Give the schedule a name and specify the necessary settings.

4. **Start the schedule**

   Select **Start schedule**.

## Trend data

To stack, overlay, and compare trends for time series, `click+drag` or `hover+scroll` to adjust and move each diagram axis.

Use the **Slider** on the toolbar to view or compare time series values at a specific time.

## View and assess data quality

Data profiling uses the **Industrial Data Science Library (IndSL)** to define gaps if the time deltas deviate by more than 1.5 times the interquartile range.

1. **Open Data profiling**

   In the chart, select **Data profiling**. Select the time series you want to assess in the **Source** dropdown list.

2. **View quality metrics**

   Switch between viewing a boxplot and a histogram of time deltas or density.

## Set thresholds for historical data

To set thresholds for historical data:

1. **Open Thresholds**

   On the top right of the chart, select the **Threshold** icon.

2. **Configure threshold values**

   In the panel, set the threshold values for time series and calculations.

## Add events

You can add events, such as a faulty shutdown, to help analyze time series.

To add events to a chart:

2 **Configure event filters**

Add or select a filter to specify the events to add.

## Add activities

Activities in Cognite Data Fusion (CDF) represent tasks, events, or operations that happen over a period of time in your industrial environment. They provide a structured way to track work, maintenance, inspections, and other time-based activities, connecting planned and executed work with your physical assets and operational data.

Turn on **Activity** to visualize and contextualize time series data with these operational events in your chart. Activities appear as vertical bars in the chart, providing visual reference to help you correlate time series behavior with relevant operational activities such as maintenance work, inspections, or production events.

### Prerequisites

Before you add activities to a chart, ensure that:

- The relevant data is modeled as **activities or extended activities in your CDF project**.
- You have time series `read` access rights to view Charts and the relevant data.

To add activities to a chart:

1 **Add time series**

Select **+ Add data** > **Add time series**. Search and select the time series you want to visualize.

2 **Select activities**

the chart. You can clear or add filters based on your preferences.

Activities appear as vertical blue bars on the **Activity** layer. These bars help you visually correlate time series with relevant activities in the selected time frame. The bars have varying levels of transparency to distinguish overlapping activities.

To work with activities:

1. **View activity details**

   Click an activity bar to view details such as start and end times, a description, and the asset the activity is tied to. Click **View details** to see full information about the activity.

2. **Pin an activity**

   Click the pin icon to keep the activity highlighted while you compare it with other activities.

## Monitor time series

Time series monitoring jobs send **emails for any breaches to the thresholds** you set. If there are several breaches within the *minimum duration period*, you'll only receive one email for the period.

We recommend that you set a minimum duration of 5 minutes to limit the number of email alerts you receive. The minimum expected data point frequency is 1 minute. If the data points have a lower frequency, alerts may not trigger as expected.

Select **Monitoring** > **Create** from the sidebar.

2 **Name the job**

Give the monitoring job a **name**.

3 **Select a source**

Select the time series or calculation you want to monitor in the **Source** dropdown.

4 **Set threshold and duration**

Specify the **threshold** and **minimum duration** for the alert.

5 **Specify storage folder**

Specify the folder to store the alerts in.

6 **Select recipients**

Select the users who'll receive the email alerts.

7 **Optional. Configure advanced options**

Open **Advanced options** and specify the sign-in credentials for the monitoring job.

8 **Start monitoring**

Select **Start monitoring**.

## Subscribe to alerts

Select **Monitoring** from the sidebar.

2 **Subscribe to alert**

Find the alert you want to subscribe to and select 🔔.

### Resolve alerts

To **resolve** an alert you've investigated:

1 **Open alert history**

Select the alert > **History**.

2 **Mark as resolved**

Select **Active** > **Mark as resolved**.

## Data aggregation

If a chart contains more than 100,000 data points, Charts summarizes the data points to make the data easier to view. The aggregated data is shaded and shows the minimum and maximum values of the aggregated data range. Zoom in to view individual data points.

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